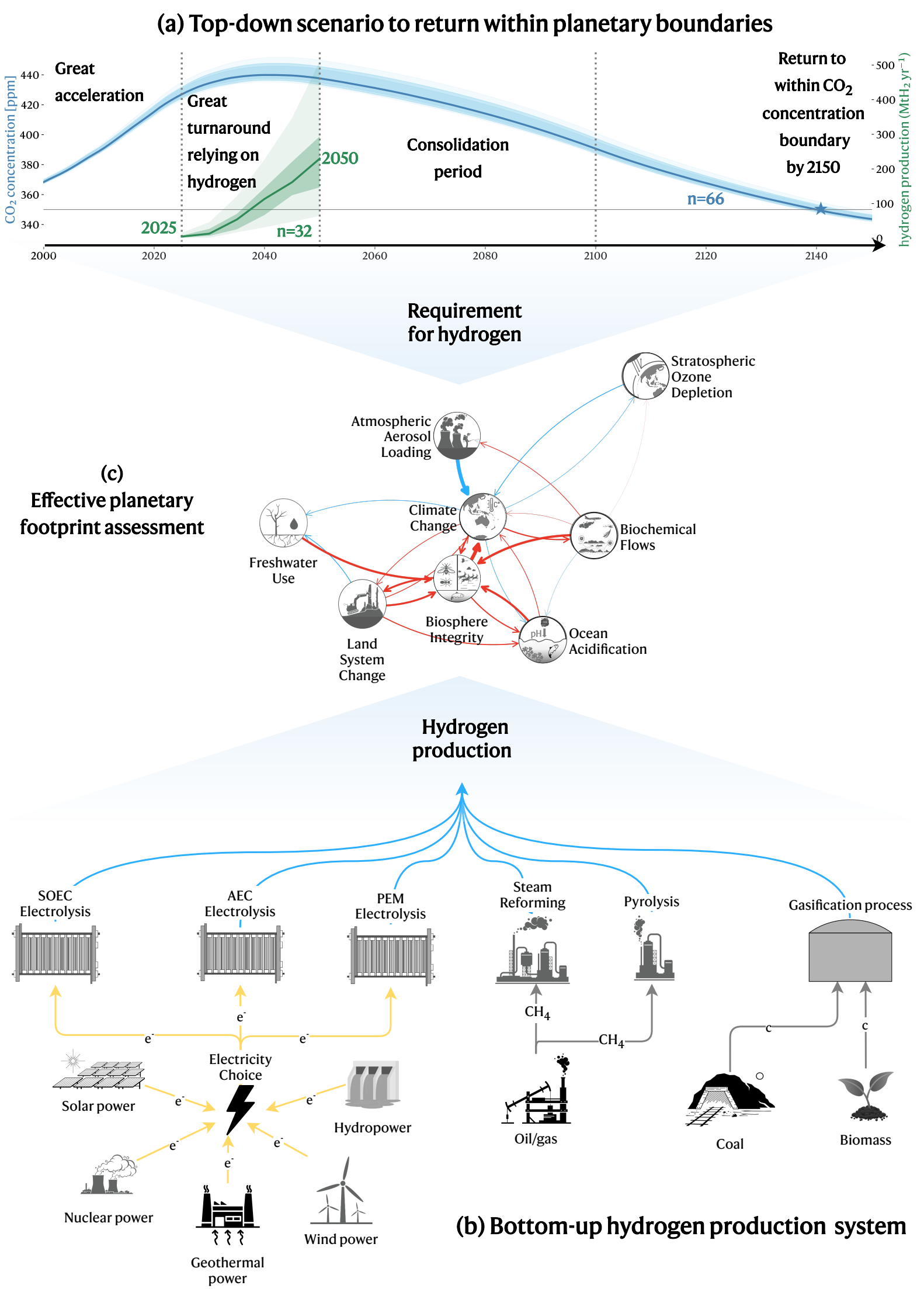
(a) Top-down scenario to return within planetary boundaries
Great acceleration
Great turnaround relying on hydrogen
Consolidation period
Return to within CO2 concentration boundary by 2150
2025
2050
n=32
n=66
CO2 concentration [ppm]
hydrogen production (MtH2 yr-1)
Requirement for hydrogen
(c) Effective planetary footprint assessment
Stratospheric Ozone Depletion
Atmospheric Aerosol Loading
Climate Change
Biochemical Flows
Freshwater Use
Biosphere Integrity
Land System Change
Ocean Acidification
Hydrogen production
SOEC Electrolysis
AEC Electrolysis
PEM Electrolysis
Steam Reforming
Pyrolysis
Gasification process
CH4
Electricity Choice
Solar power
Hydropower
Nuclear power
Geothermal power
Wind power
Oil/gas
Coal
Biomass
(b) Bottom-up hydrogen production system